National University of Computer and Emerging Sciences
Karachi Campus

Theory Of Automata

Sessional-II Exam

Date: November 6th 2025

Total Time (Hrs): 01

Course Instructor(s)

Total Marks: 30

Total Questions: 02

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Roll No Section Student Signature

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INSTRUCTION:

- Return the question paper.
- Read each question completely before answering it. There are 3 **questions on 1 page 2 sides**.
- In case of any ambiguity, you may make assumptions. But your assumption should not contradict any statement in the question paper.
- Start each question in a new page.

CLO #2: Prove properties of languages, grammars and automata with rigorously formal mathematical methods.

Question 1a: Kleene's Theorem: **[10 mins, 6 Points]**

Find the Concatenation of FA1 and FA2, given in figure 1 and figure 2, and the closure of FA2 given in figure 2.

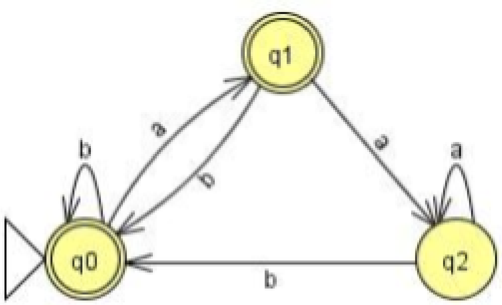


figure 1: FA1

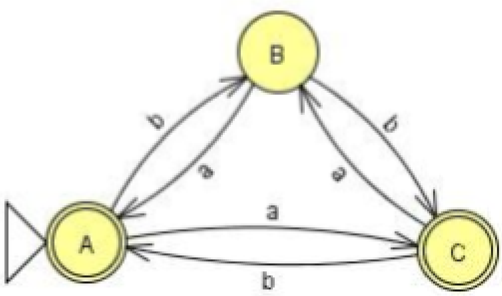


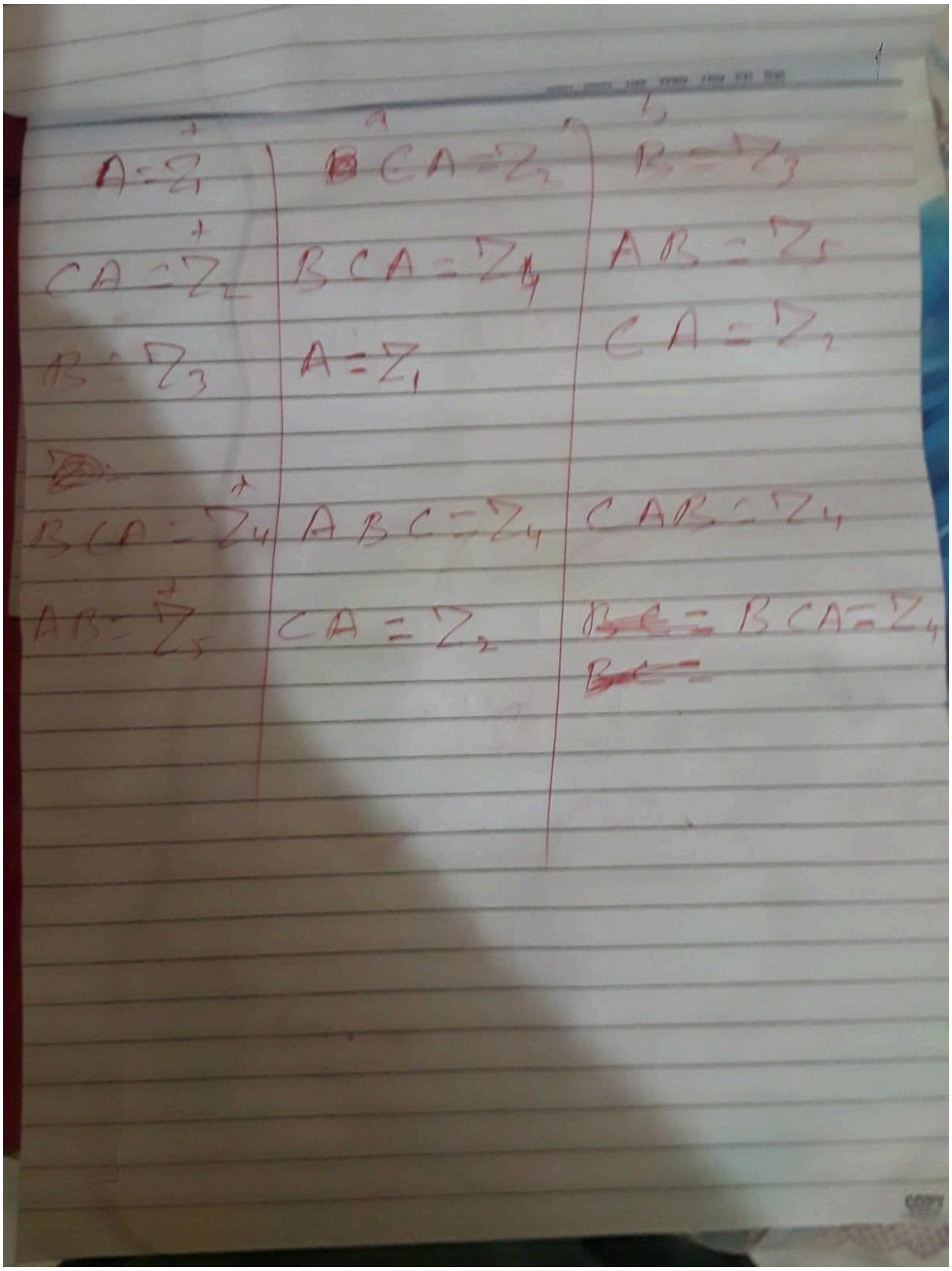
figure 2: FA2

Concatenation

	a	b
$q_0 A = Z_1$	$q_1 AC = Z_2$	$q_0 AB = Z_3$
$q_1 AC = Z_2$	$q_2 CB = Z_4$	$q_0 AB = Z_3$
$\cancel{q_2 CB = Z_4}$	$q_2 BA = Z_5$	$q_0 AC = Z_7$
$q_0 AB = Z_3$	$q_1 AC = Z_2$	$q_0 ABC = Z_5$
$q_2 CB = Z_4$	$q_2 BA = Z_6$	$q_0 AC = Z_7$
$q_0 ABC = Z_5$	$q_1 ACB = Z_8$	$q_0 ABC = Z_5$
$q_2 BA = Z_6$	$q_2 AC = Z_9$	$q_0 ACB = Z_5$
$\cancel{q_2 AC = Z_7}$		
$q_0 AC = Z_7$	$q_1 ACB = Z_8$	$q_0 AB = Z_3$
q [illegible] $= Z_8$	$q_2 CBA = Z_{10}$	$q_0 ABC = Z_5$

$q_2 AC = Z_9$	$q_2 CB = Z_4$	$q_0 AB = Z_5$
$q_2 CBA = Z_{10}$	$q_2 BAC = Z_{10}$	$q_0 ACB = Z_6$

closure



	a	b
$A = \Sigma_1$	$CA = \Sigma_2$	$B = \Sigma_3$
$CA = \Sigma_2$	$BCA = \Sigma_4$	$AB = \Sigma_5$
$B = \Sigma_3$	$A = \Sigma_1$	$CA = \Sigma_2$
$BCA = \Sigma_4$	$ABC = \Sigma_4$	$CAB = \Sigma_4$
$AB = \Sigma_5$	$CA = \Sigma_2$	$\cancel{BC} = BCA = \Sigma_4$

Question-1b: Grammars: **[20 mins, 3+3+3+3+2 Points]**

Given the following Language, identify whether the language is a regular or context-free language, then express the respective grammar. For regular language, create a Regular Grammar, or for CFL create CFG: Convert RG of part 2 into Equivalent FA

1. $(u^{2n+3})(v^n)$; n > 0

CFG

S → uu S v | uuu

2. L1={w ∈ {a,b}*|w starts and ends with the same symbol}

RG

$S \rightarrow aA \mid bB \mid a \mid b$

$A \rightarrow aA \mid bA \mid a$

$B \rightarrow aB \mid bB \mid b$

3. L2 = {w: $|w|_a = 3 \bmod 5$}, where $\sum$ ={a,b}

   RG

   $S0 \rightarrow a\ S1 \mid b\ S0$

   $S1 \rightarrow a\ S2 \mid b\ S1$

   $S2 \rightarrow a\ S3 \mid b\ S2$

   $S3 \rightarrow a\ S4 \mid b\ S3 \mid \varepsilon$

   $S4 \rightarrow a\ S0 \mid b\ S4$

   For example:aaa, aabbbaabbbaaabbba,ababababababababababaaa

4. All strings with exactly two a's and more than two b's.

RG

$S0 \rightarrow b\ S1 \quad \mid a\ A0$

$S1 \rightarrow b\ S2 \quad \mid a\ A1$

$S2 \rightarrow b\ S3 \quad \mid a\ A2$

$S3 \rightarrow b\ S3 \quad \mid a\ A3$

$A0 \rightarrow b\ A1 \quad \mid a\ B0$

$A1 \rightarrow b\ A2 \quad \mid a\ B1$

$A2 \rightarrow b\ A3 \quad \mid a\ B2$

$A3 \rightarrow b\ A3 \quad \mid a\ B3$

$B0 \rightarrow b\ B1$

$B1 \rightarrow b\ B2$

$B2 \rightarrow b\ B3$

$B3 \rightarrow b\ B3 \mid \varepsilon$

5. L = { $a^n b^m c^k \mid n = m \text{ or } m \leq k$ }

   CFG

   $S \rightarrow S_1C \mid AS_2$

   $S_1 \rightarrow aS_1b \mid \varepsilon$

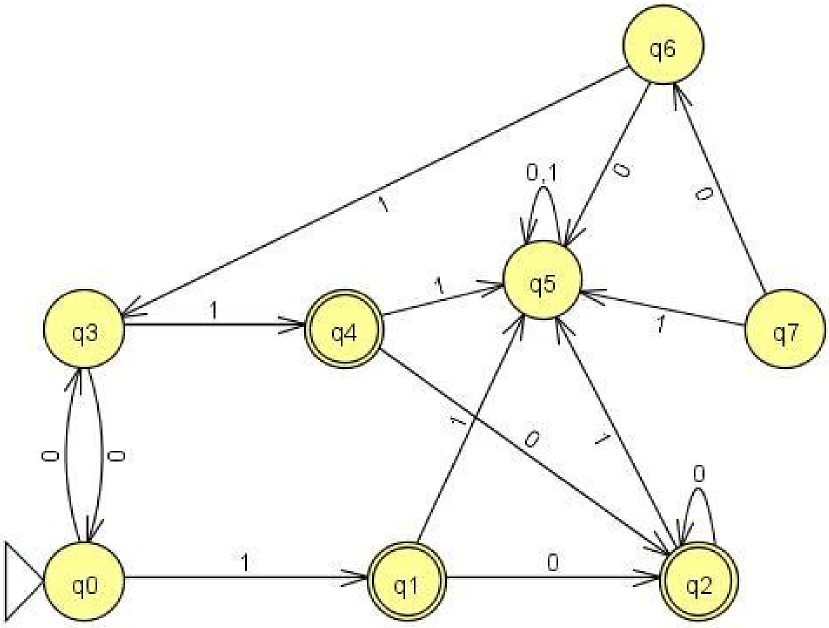
   $S_2 \rightarrow bS_2c \mid C$

   $C \rightarrow cC \mid \varepsilon$

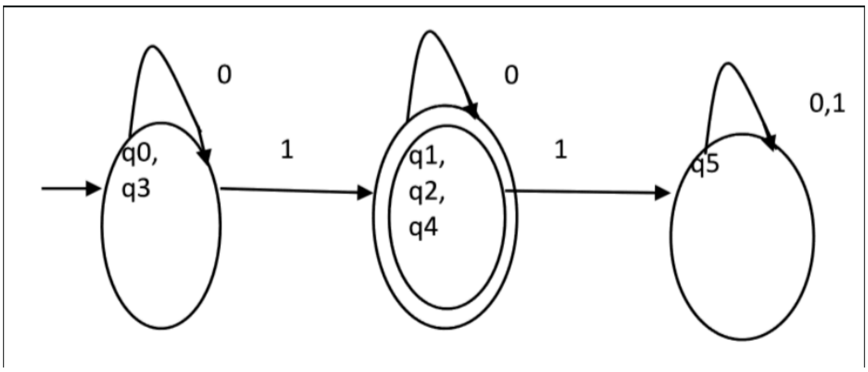
   $A \rightarrow aA \mid \varepsilon$

Question 2a: **DFA Minimization:** **[30 mins, 5+5 Points]**

Given the following DFA M. apply the DFA minimization algorithm and show the resulting minimized DFA. Show your graph or table and label each state in your minimal DFA with the set of states of M it corresponds to.



Solution



Question-2b Simplification:

S ⟶ ASA | AB | BCD | DFG | EFG | AFG

A ⟶ Abc | BCa | aAc |ε

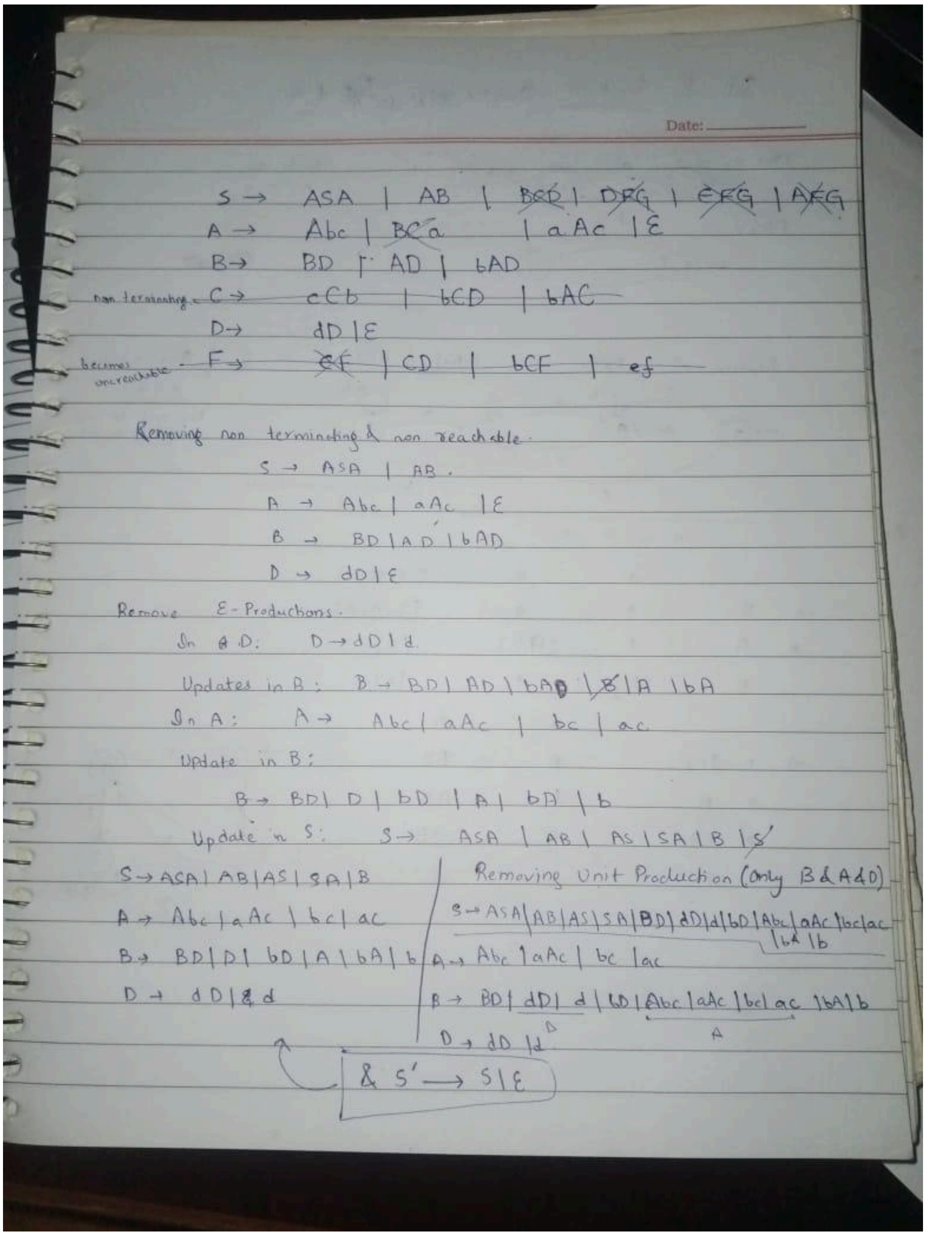
B ⟶ BD | AD | bAD

C ⟶ cCb | bCD | bAC

D ⟶ dD | ε

F ⟶ EF | CD | bCF | ef

Simplify the grammar and show all the steps.



Date:

$S \rightarrow ASA \mid AB \mid \cancel{BCD} \mid \cancel{DFG} \mid \cancel{EFG} \mid \cancel{AEG}$

$A \rightarrow Abc \mid \cancel{BCa} \mid aAc \mid \varepsilon$

$B \rightarrow BD \mid AD \mid bAD$

non terminating – $\cancel{C \rightarrow cCb \mid bCD \mid bAC}$

$D \rightarrow dD \mid \varepsilon$

becomes unreachable – $\cancel{F \rightarrow EF \mid CD \mid bCF \mid ef}$

Removing non terminating & non reachable.

$S \rightarrow ASA \mid AB$.

$A \rightarrow Abc \mid aAc \mid \varepsilon$

$B \rightarrow BD \mid AD \mid bAD$

$D \rightarrow dD \mid \varepsilon$

Remove ε-Productions.

In D: $D \rightarrow dD \mid d$.

Updates in B: $B \rightarrow BD \mid AD \mid bAD \mid \cancel{B} \mid A \mid bA$

In A: $A \rightarrow Abc \mid aAc \mid bc \mid ac$

Update in B:

$B \rightarrow BD \mid D \mid bD \mid A \mid bA \mid b$

Update in S: $S \rightarrow ASA \mid AB \mid AS \mid SA \mid B \mid \cancel{S}$

$S \rightarrow ASA \mid AB \mid AS \mid SA \mid B$

$A \rightarrow Abc \mid aAc \mid bc \mid ac$

$B \rightarrow BD \mid D \mid bD \mid A \mid bA \mid b$

$D \rightarrow dD \mid d$

$\boxed{\& \; S' \longrightarrow S \mid \varepsilon}$

Removing Unit Production (Only B & A & D)

$S \rightarrow ASA \mid AB \mid AS \mid SA \mid BD \mid dD \mid d \mid bD \mid Abc \mid aAc \mid bc \mid ac \mid bA \mid b$

$A \rightarrow Abc \mid aAc \mid bc \mid ac$

$B \rightarrow BD \mid \underbrace{dD \mid d}_{D} \mid bD \mid \underbrace{Abc \mid aAc \mid bc \mid ac}_{A} \mid bA \mid b$

$D \rightarrow dD \mid d$

========================== Best of Luck ==========================